

Promoting an efficient and eco-friendly solution: A Carpooling strategy for Rutgers University

Rutgers University faces the issues of transportation, high campus traffic, and the rising environmental concerns. Currently, there are 19.4 million college students in the US, and 51% of them do not have access to a car on campus (Gavrilenko, 2022). In addition, Rutgers faces the problem of limited parking availability, and single-occupied cars have continually risen the total carbon emissions. This causes an issue specifically for commuters, staff, or those who live in off-campus housing, as transportation to other campuses have been increasingly difficult. To address this situation, a carpooling strategy can be implemented by Rutgers, to make transportation more efficient and overall cause less campus traffic.

This problem specifically focuses on students who live off-campus, staff, and commuters. There are currently over 43,000 students in Rutgers New Brunswick (Rutgers, 2024). Not including the staff, which if added on, further highlights the problem of transportation and finding available parking. As for most faculty, their main source of transportation are their cars, so prioritizing parking spaces for staff is the first thing Rutgers implements. This only causes a temporary solution, as there are much more students than faculty, so parking is already significantly occupied. In order to address this problem, implementing a Rutgers carpooling strategy that caters to both students and faculty, can potentially create a smoother transportation system. In turn, this can solve Rutgers traffic congestion, while also promoting an environment friendly community that can thrive off of helping each other.

Funders for this initiative would be Rutgers University, or other companies that are working towards solving problems towards campus traffic. Locally speaking, Rutgers can help design and fund a project with the help and effort of other students, to create a carpooling software that specifically focuses on Rutgers community. With only needing a small budget, Rutgers could potentially save more money in implementing a carpooling system, than further spending the budget on creating more parking spaces. Additionally, funding can also come from outside sponsors or alumni, who have faced similar issues in the past. Using other resources like gofundme can help towards creating the small budget needed to implement this system. This way, the project turns into realistic expectations, with only needing combined efforts from the community and university, to address the constant problems of campus congestion.

The stakeholders in this project are mainly the faculty and students. More specifically, students who commute or live off-campus. A similar survey has been conducted by another group, and their response was similar. It states, "The results showed that most of the surveyed respondents are oriented towards using carpooling services for their educational (university/school) purposes, meaning carpooling will be more attractive for people like students." (Li, 2023). This specific population in Rutgers are mostly the ones who use their car for daily transportation, and being able to address this issue can alleviate stress throughout the whole university. Commuters or other students can share rides, potentially improving the issue of parking availability. Similarly, faculty and staff can share the same process, which can also reduce the amount of occupied parking spaces in faculty lots. Not only does this increase parking availability, it also reduces campus congestion, with less people needing to take other transportation options like the buses

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or walking. With fewer people taking the buses, the need for a new or revised transportation system is potentially solved, just by implementing a Rutgers carpooling system.

The methodology behind the Rutgers carpooling system can be very simple, keeping it friendly for all potential users. One method used by an app called PoliUniPool, has a similar process in how to approach Universities with a carpooling system. It states, “The main characteristics of the PoliUniPool service are the following: (1) the use of the system is restricted to employees, faculty and students of the two universities; (2) besides suggesting a matching between the users, the system provides the expected schedule for their trips; (3) in addition to the campus premises, users can select – as destination of their car pooling trips – the main railway and subway stations, in order to encourage the most environmental friendly means; (4) users are informed immediately in case of delay or changes, to improve the reliability of the service; (5) the system estimates the costs for each user, in order to let the users know how to share them; (6) the system has some social network functionalities, e.g. drivers are able to set partial pre-arranged crews; and users may indicate other users they would prefer to car-pool with (“friends”) or they don’t want to (“I don’t like him/her”).”(Bruglieri, 2011). This method highly focuses on a user-friendly software that aids students and faculty to a safe and reliable source of transportation. Using methods such as surveys or forms, can extract the information on the overall need or problems that students or faculty face, when it comes to campus congestion and parking availability. This information will help towards specifying which campus or demographic faces the most of these issues, which can then narrow down where efforts need to be focused. The next step will be how to implement the Rutgers carpooling system.

Previous methods in addressing this issue are creating an app, or a social media group. Although, there are a few problems that arise when approaching these methods. In Li’s article it states, “Attitudes towards carpooling suggest that it is a familiar concept or that college students believe that it is possible but just not yet widely used among college students. Many respondents said that the current market in carpooling is not formal enough, security is not guaranteed and more than one carpool will increase travel time. Among the respondents, the top two concerning factors of carpooling are travel costs and security.”(Li, 2023). With this in mind, a solution created officially through Rutgers can help address the problems of safety and travel costs. Implementing a specific widget in myRutgers Portal can make the carpooling transition convenient and simple. This way, the carpooling system is catered only towards Rutgers students and faculty, reducing the risk of unknown users. To add, being able to post your schedule in this widget can help narrow down the issues of finding someone compatible with your schedule. Lastly, being able to give feedback on the rides will be able to connect students and faculty, or to address what they didn’t like about the driver or experience. This gives the users freedom to be able to choose who they want to ride with, or the option to ride with somebody else.

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